

Phase I/II Statistical Process Monitoring SECOM Data

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統計專論 (III) Final Report

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Project Objective

- Dataset: UCI SECOM semiconductor manufacturing data.
- Objective: construct a complete SPC workflow.

Phase I diagnosis → control chart design → Phase II monitoring

- Main comparison: Shewhart, EWMA, CUSUM, and CPD charts.
- Final question: which chart signals first, and whether the pattern suggests mean shift, variance shift, or structural change.

Selected Variable and Preprocessing

Variable

Selected process variable: **x297**.

- Reason for selection: low missing rate, high number of distinct values, and sufficient process variation.
- Raw observations: 1567.
- Missing observations removed: 2.
- Outlier rule: global 1.5 IQR filter after dropping missing values.
- Outliers removed: 128.
- Cleaned observations: 1437.

Variable Summary

Mean	SD	Min	Q1	Median	Q3	Max
1410.91	1126.51	0.00	556.99	1061.44	1995.39	4945.92

- The cleaned variable remains right-skewed with large dispersion.
- Since this is an individual process variable, all charts are built for individual observations.

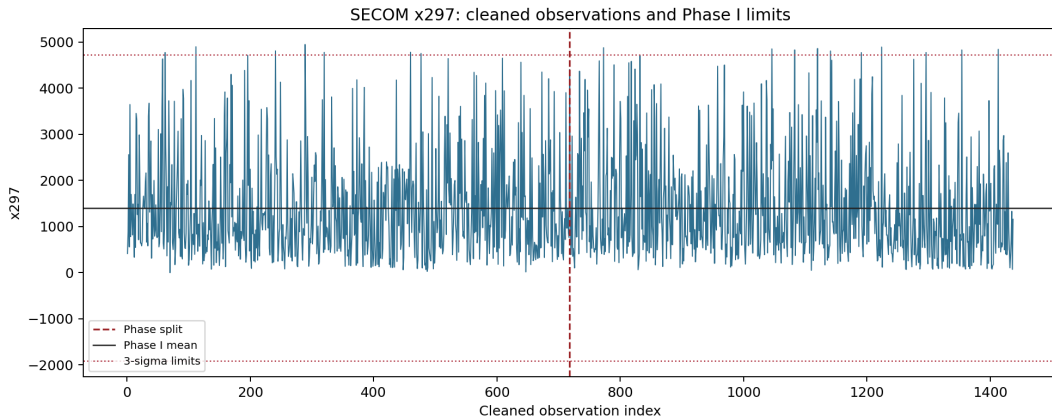
Phase I / Phase II Split

$$m = \lfloor 0.5n \rfloor = \lfloor 0.5(1437) \rfloor = 718.$$

Phase	Observations	Mean / SD
Phase I	718	1403.64 / 1105.53
Phase II	719	1418.18 / 1147.80

- Phase I is used as the in-control reference sample.
- Phase II is monitored sequentially using the chart limits estimated from Phase I.

Cleaned Series and Phase Split



Recursive segmentation + permutation result

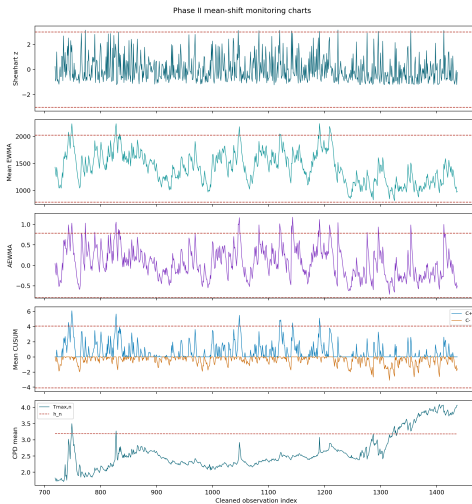
$$p_{\text{level}} = 0.8675, \quad p_{\text{scale}} = 0.8325.$$

- Both p-values are much larger than 0.05.
- Phase I shows no strong evidence of level or scale instability after preprocessing.
- The Phase I sample is therefore treated as the IC reference for Phase II charts.

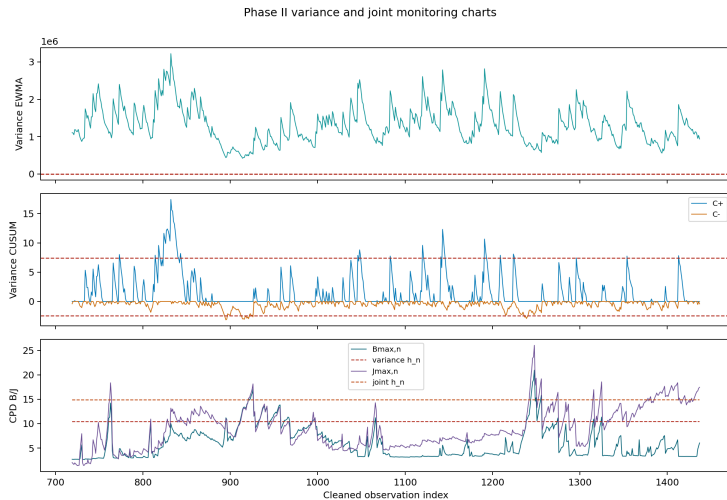
Method	Target	Main setting
Individuals Shewhart	Mean	3-sigma limits
Mean EWMA	Mean	$\lambda = 0.1$, $\rho = 2.454$
Variance EWMA	Variance	$\lambda = 0.1$, $\rho_U = 2.595$, $\rho_L = 1.580$
Adaptive EWMA	Mean	eta1 case (ii), $h = 0.7874$
Mean CUSUM	Mean	$k = 0.5$, $h = 4.095$
Variance CUSUM	Variance	$k_U = 1.848$, $h_U = 7.416$; $k_L = 0.462$, $h_L = -$
CPD charts	Mean / variance / joint	$\alpha = 0.005$

- EWMA and CUSUM settings follow the in-class R code.
- CPD limits follow Chapter 6 approximations for mean, variance, and joint charts.

Mean-Shift Monitoring Results



Variance and Joint Monitoring Results



Signal Comparison

Chart	First signal	Signal count	Target
Variance EWMA	719	719	variance
Mean EWMA	743	17	mean
Adaptive EWMA	743	32	mean
Mean CUSUM	743	13	mean
CPD mean chart	749	118	mean
CPD variance chart	761	78	variance
CPD joint chart	762	71	mean/variance
Individuals Shewhart	773	10	mean
Variance CUSUM	773	59	variance

- Earliest signal: variance EWMA at cleaned index 719, the first Phase II observation.
- Among mean-sensitive charts, EWMA, AEWMA, and CUSUM signal first at index 743.

CPD Post-Signal Interpretation

- CPD mean chart first signals at index 749.
- CPD variance chart first signals at index 761.
- CPD joint chart first signals at index 762.
- The strongest CPD variance and joint signals estimate the likely change location around:

$$\hat{r} \approx 1229.$$

- The strongest CPD mean signal estimates:

$$\hat{r} \approx 1212.$$

- Phase I RS/P diagnosis does not reject stability, so the reference sample is usable.
- Phase II mean changes are detected by EWMA, AEWMA, CUSUM, and CPD mean chart.
- Variance-sensitive methods are more aggressive, especially variance EWMA.
- CPD variance and joint charts locate later structural changes around indices 1229.
- Overall pattern suggests not a single isolated mean shift only, but a combination of mean instability and variance/structural change.

Conclusion

- ① The selected SECOM variable x297 is usable after missing-value removal and IQR outlier filtering.
- ② Phase I RS/P results support using the first 718 cleaned observations as the IC reference.
- ③ The earliest signal is variance EWMA at the first Phase II observation.
- ④ For mean-shift detection, mean EWMA, AEWMA, and mean CUSUM signal earliest at index 743.
- ⑤ CPD charts provide additional localization and suggest later structural changes around indices 1212–1229.